

CYBERIUM-CLI

TERMINAL-FIRST OFFENSIVE SECURITY LEARNING SUITE

A native Linux Proof of Concept eliminating browser context switching and streamlining student progression during offensive cyber labs.

AUTHOR:
Denis Albin

STUDENT CODE:
S2

INSTITUTION:
John Bryce Academy

ARCHITECTURE:
Pure Python 3 (0 pip)

TARGET LAB:
Metasploitable 2 (Linux Lab)

TEST SUITE:
34/34 Passing (100%)

0	< 50ms	100%	14
PIP DEPENDENCIES	API SOLVE LATENCY	TERMINAL NATIVE	ENROLLED PROGRAMS

AUTHOR'S STATEMENT: THE PHILOSOPHY OF "LIVING IN THE TERMINAL"

As an offensive security student at John Bryce Academy, I fundamentally love **"living in the terminal"**. That is where real penetration testing happens—crafting payloads, automating scans, analyzing responses, and chaining exploits.

During intensive multi-hour lab sessions, having to constantly tab-switch between a Kali Linux terminal and an external web browser introduces recurring friction. Every time you leave the shell to check a question prompt, verify a target parameter, or submit a flag, your flow state breaks.

I developed `cyberium-cli` out of this exact personal experience: to bring syllabus objectives, instant flag submission, course materials, and field scratchpad notes directly into the shell. While others may prefer working in a web browser, **working via this terminal tool drastically increased my personal focus, learning velocity, and problem-solving speed**. At the same time, I engineered the tool to be universal and lightweight for any student—running out of the box with zero external dependencies across VMware Workstation, VirtualBox, native Linux, or SSH jump boxes.

[✓] FULL LAB AUTONOMY

Submit flags, download targets, view course books, and inspect progress tables without leaving the terminal shell.

[✓] UNIVERSAL LAB COMPATIBILITY

Operates seamlessly across standard student environments: Kali on VMware, VirtualBox, native Linux, WSL2, or remote SSH jump hosts.

PURE PYTHON 3 STANDARD LIBRARY ARCHITECTURE (ZERO PIP)

To ensure total reliability across diverse student machines—including restricted lab images, corporate laptops, or offline VMs without pip access—`cyberium-cli` was engineered with **strictly zero external pip dependencies**.

urllib.request
Native HTTP/S client with keep-alive and Cloudflare load-balancing header injection.

curses & sqlite3
Full terminal GUI engine with zero ncurses bindings, plus direct browser cookie DB parsing.

json & subprocess
Atomic configuration persistence, clipboard sync, and instant PDF course book launch.

STREAMLINED STUDENT AUTHENTICATION & ONBOARDING

To make lab onboarding fast and frictionless for any student, `cyberium-cli` provides two simple authentication workflows without requiring manual API or network configuration:

- **Automatic Browser Sync (`cyberium sync`):** Automatically extracts the active student session from Firefox, Chrome, Chromium, or Brave databases on the local machine in under 1 second.
- **Direct Terminal Login (`cyberium login`):** Designed for headless virtual machines, VMware instances without a GUI browser, or remote SSH sessions—prompts the student to paste their session token once to link the account. Credentials persist securely in `~/.config/cyberium/config.json`.

kali@box: ~ (SSH Remote Kali Onboarding Flow & Module Completion) [✓] AUTHENTICATED

```
❯ ssh

┌─ 14:14  ~
└─ kali ❱ cyberium

[×] No Cyberium student session detected.
Each student needs their own session cookies to submit answers.

Would you like to log in now? [Y/n]: y

C Y B E R I U M [JOHN BRYCE] | STUDENT AUTHENTICATION
──────────────────────────────────────────────────────
Each student has their own unique session on Cyberium Academy.
Follow these steps to link your terminal client manually:

  1. Log into https://cyberium.academy in your browser.
  2. Press F12 → Storage / Application → Cookies.
  3. Copy the value for cyb_rt (JWT refresh token).

Enter cyb_rt: eyJhbGciOi... [REDACTED_STUDENT_SESSION_JWT]

Enter __cflb (optional, press Enter to skip): [REDACTED_CLOUDFLARE_ROUTING_KEY]
Enter connect.sid (legacy / optional, press Enter to skip):
Enter cyb_td (optional, press Enter to skip):

[✓] Credentials saved securely to /home/kali/.config/cyberium/config.json!
Your session is independent to your local user account.


C Y B E R I U M [JOHN BRYCE] | Post (Linux) (Progress: 83/83)
──────────────────────────────────────────────────────
  Completion: [████████████████████████████] 100%

  [✓] All 83 lab questions in this module are COMPLETED!


┌─ 14:16  ~   1m11s
└─ kali ❱ cyberium
```

FIGURE 1: Universal terminal onboarding flow, zero-configuration credential caching, and module completion verification.

SELF-DOCUMENTING CLI ARCHITECTURE & BUILT-IN COMMAND REFERENCE

A core objective of `cyberium-cli` is total self-containment. Rather than requiring students to search web portals or documentation manuals, the CLI provides complete self-documenting inline help (`cyberium -h`). Below illustrates the full positional command tree and practical operational examples:

kali@box: ~ (cyberium -h: Positional Commands & Core Arguments) COMMAND REFERENCE: PART 1

```
16:09 ~
kali ❯ cyberium -h
usage: cyberium [-h]
                {status,current,show,list,modules,set-module,module,programs,tracks,set-program,program,switch,track,profile,stats
,banner,book,video,extra,themes,set-theme,theme,search,note,notes,login,logout,sync,refresh,tui,submit,submit-extra} ...

Cyberium Academy Terminal Client & Submission Tool

positional arguments:
  {status,current,show,list,modules,set-module,module,programs,tracks,set-program,program,switch,track,profile,stats,banner,book,v
ideo,extra,themes,set-theme,theme,search,note,notes,login,logout,sync,refresh,tui,submit,submit-extra}
    status (current, show)
                        Show current question & progress (default)
    list                List all questions in current module
    modules             List all modules in current program
    set-module (module)
                        Set or choose active module
    programs (tracks)   List all enrolled programs
    set-program (program, switch, track)
                        Switch active track/program (interactive if empty)
    profile             Show student profile & score
    stats               Full student operator dashboard & rank
    banner              Display Cyberium ASCII banner & stats
    book                Get course PDF book, download, or open
    video               Get HD video lecture streaming link
    extra               List bonus/extra challenges
    themes              List all available terminal themes
    set-theme (theme)   Set active terminal theme
    search              Search questions by keyword
    note (notes)        Add, view, or list question scratchpad notes
    login               Interactive login / session sync
```

kali@box: ~ (cyberium -h: Practical Usage Examples & Subcommand Syntax) COMMAND REFERENCE: PART 2

```
Apps  Places  Wed Oct 14  7:10 PM
❯ ssh
    submit-extra        Submit answer to extra/bonus challenge

options:
  -h, --help            show this help message and exit

Examples:
  cyberium                      # Show current active question
  cyberium list                 # List all questions and statuses
  cyberium submit "grep -rl"    # Submit answer to current question
  cyberium submit-extra 1 "ans" # Submit answer to bonus/extra challenge #1
  cyberium switch ZX301         # Switch to Penetration Testing track
  cyberium switch               # Interactive track selector
  cyberium set-module           # Interactive module selector
  cyberium modules              # List modules in active program
  cyberium programs             # List all 14 enrolled programs
  cyberium extra                # View bonus / extra challenges
  cyberium themes               # List 16 color palettes
  cyberium set-theme matrix     # Switch to matrix / classic theme
  cyberium stats                # Full operator rank, progress & stats
  cyberium note "memo"          # Save note to current active question
  cyberium note 2.2 "memo"      # Save note to question 2.2 (or #14)
  cyberium note 2.2             # View note for question 2.2
  cyberium notes                # View all saved notes across the module
  cyberium search "wget"        # Search questions by keyword
  cyberium book                 # View course PDF book link
  cyberium video                # View HD video lecture link
  cyberium profile              # View student dossier & points
  cyberium tui                  # Launch full-screen interactive TUI
  cyberium login                # Update session cookies

16:10 ~
kali ❯
```

FIGURE 2: Complete in-terminal help manual displaying supported positional commands, track management, notes, and invocation syntax.

LAB VM DEPLOYMENT: METASPLOITABLE 2 TARGET ENVIRONMENT

To demonstrate the complete student workflow, the standard John Bryce / Cyberium vulnerable target machine (**Metasploitable 2**) was deployed on an isolated student lab subnet. Whether targets are hosted locally inside VMware Workstation, VirtualBox, or on a dedicated lab bridge, the CLI client functions consistently across all environments.

TARGET NETWORK DISCOVERY & LOCAL HOST RESOLUTION

In typical student labs, vulnerable targets receive dynamic DHCP addresses (in this lab demonstration: `172.30.30.124`). Once the target IP is identified on the lab subnet, mapping it to a local alias (such as `metasploit-2` in `/etc/hosts`) enables clean, reproducible commands across all offensive tools. `cyberium-cli` displays active objectives right alongside the shell workspace.

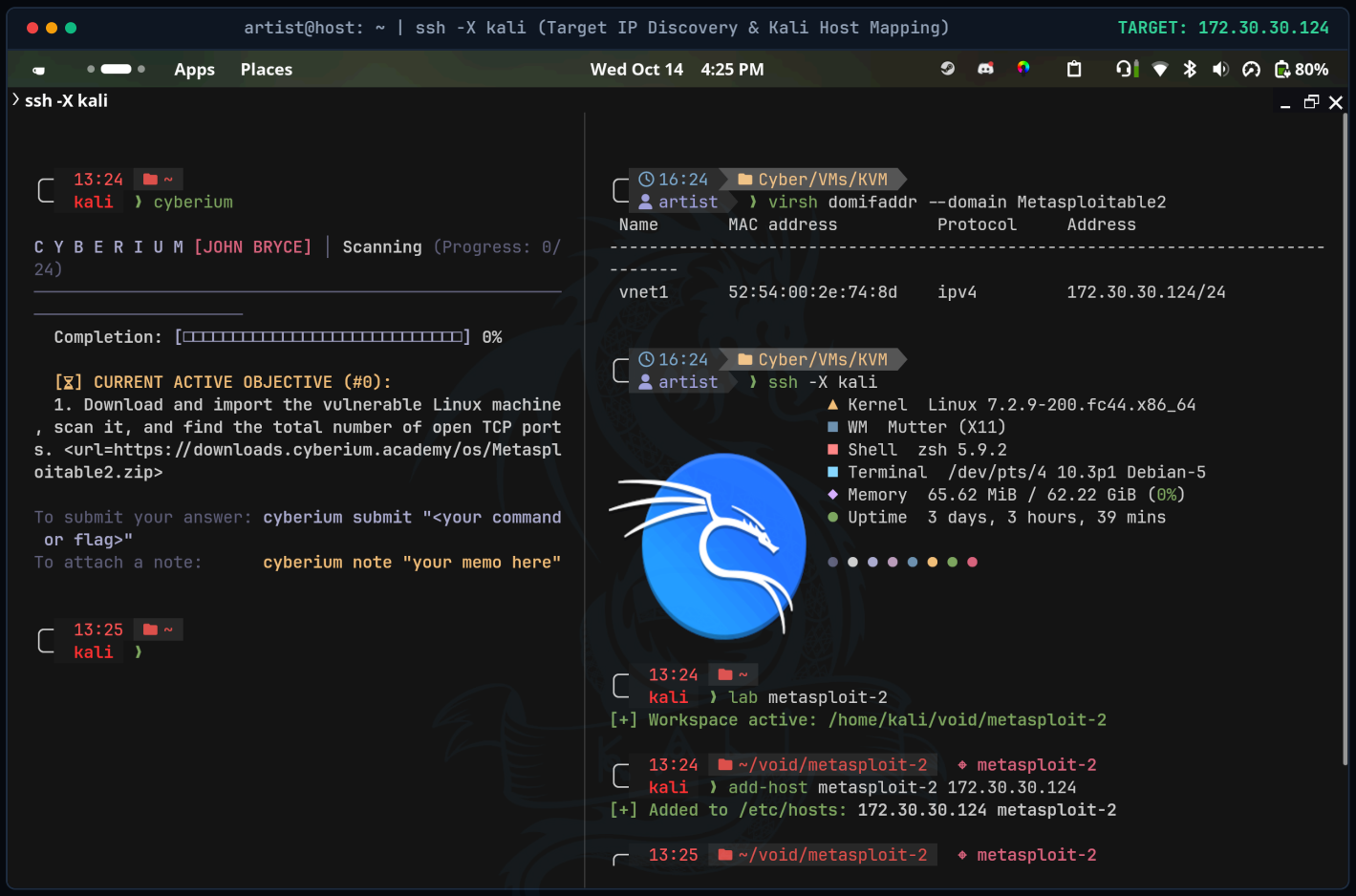
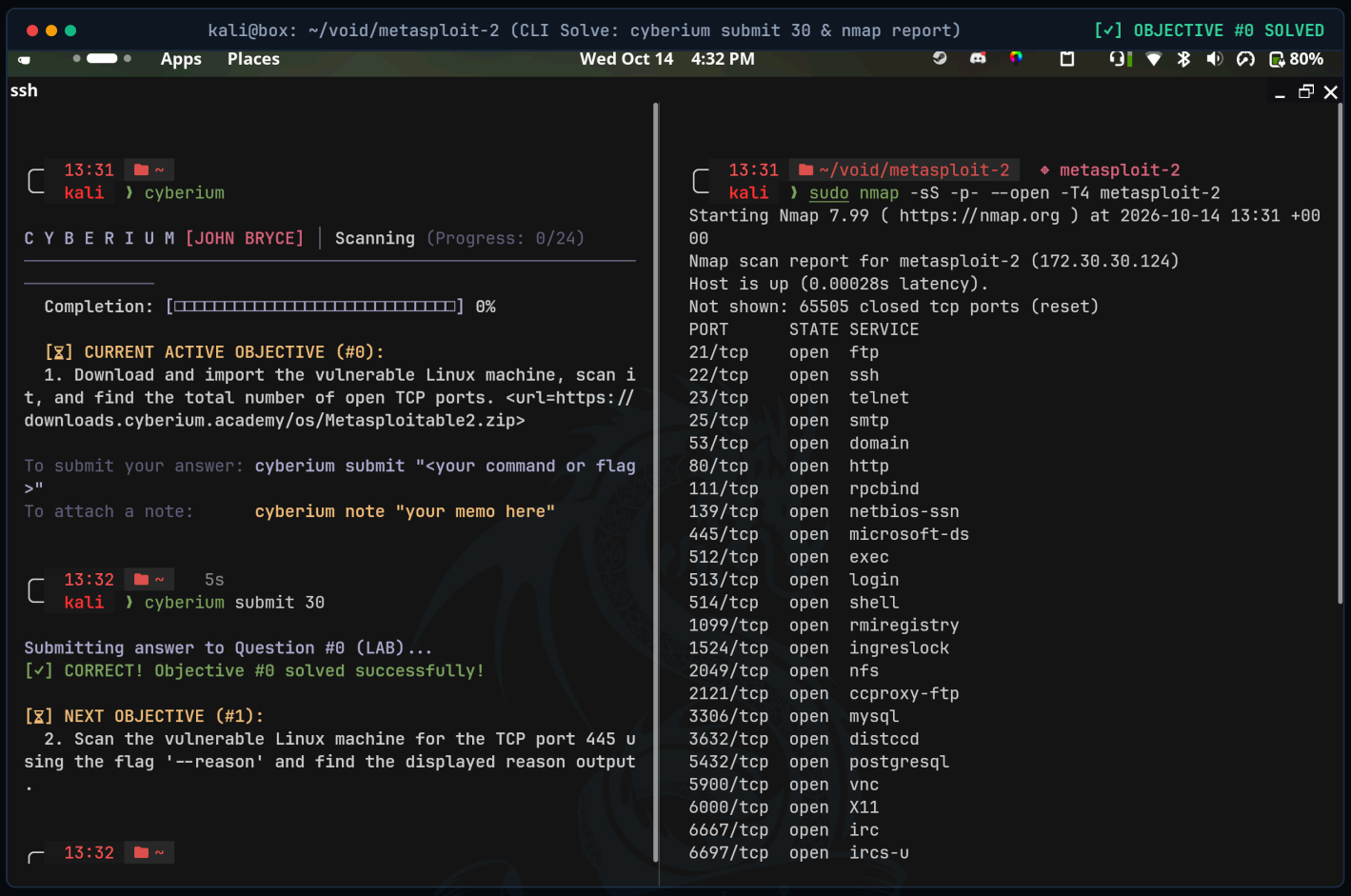


FIGURE 3: Target network discovery, host alias mapping in Kali Linux, and in-terminal active objective display.

WORKFLOW: FULL TCP PORT RECONNAISSANCE & INSTANT CLI SUBMISSION

Lab Objective #0: Scan the Metasploitable 2 target and find the total number of open TCP ports. Inside the Kali terminal, an aggressive SYN stealth scan was executed: `sudo nmap -sS -p- --open -T4 metasploit-2`, discovering exactly 30 open TCP ports. The student submits the answer instantly via `cyberium submit 30`.

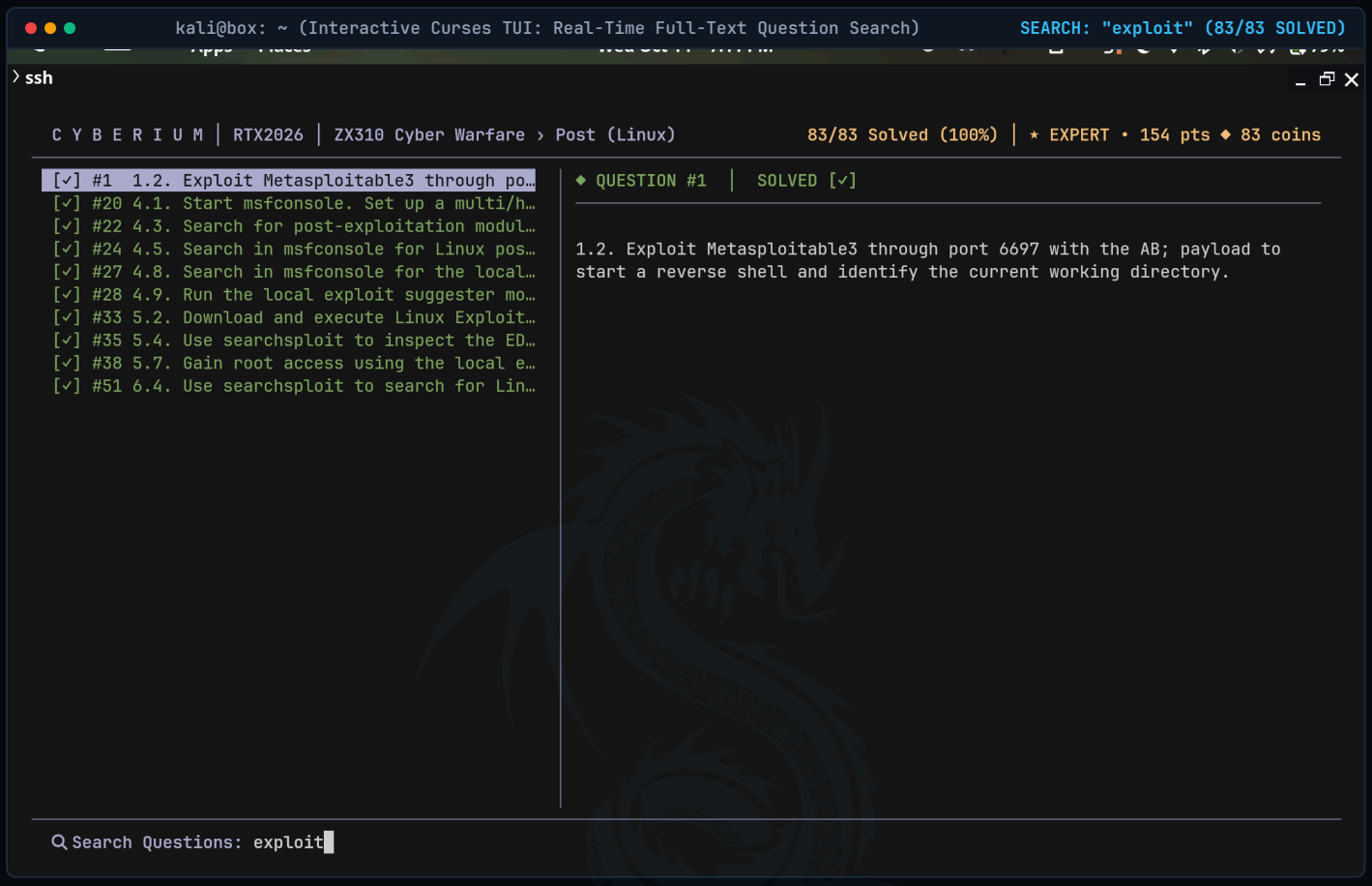


PEDAGOGICAL VALUE: REAL-TIME FEEDBACK WITHOUT CONTEXT SWITCHING

Notice how the student solved the lab objective in **5 seconds** after the port scan completed, receiving immediate green feedback and unlocking Objective #1. The student never left their Kali terminal workspace, maintaining an uninterrupted hands-on learning flow.

FULL-SCREEN KEYBOARD-DRIVEN TUI (TOKYO NIGHT PALETTE)

For students who prefer a persistent graphical cockpit over one-off commands, `cyberium tui` launches an interactive curses terminal UI styled with the Tokyo Night palette. The interface features a dual-pane layout: the left pane lists questions with instant status badges, while the right pane renders question prompts, hints, target metadata, and submission fields.



INSTANT FUZZY QUESTION FILTERING & HOTKEY CONTROLS

In comprehensive modules containing 80+ objectives, manually scrolling through lists slows down workflow. By pressing `/`, students invoke an instant full-text search bar at the bottom. As demonstrated above, filtering for `"exploit"` instantly narrows the syllabus down to matching exploitation tasks across the target machine.

[/] Live Search
Instant fuzzy filtering by keywords, tool names, or flags.

[Enter] / [n] Notes
Answer objectives directly or attach field notes to questions.

[m] / [P] / [t] Switch
Switch modules, courses, and toggle between 16 terminal themes.

MULTI-PANE ERGONOMICS & LIVE LAB CURRICULUM SYNC

Notice how the curses dashboard visualizes the entire **Post (Linux)** curriculum (83/83 completed objectives). Instead of clicking through slow web menus, students toggle between questions, read instructions, attach scratchpad notes, and switch courses with single-key shortcuts. Everything updates synchronously with the backend API.

WORKFLOW 2: INTERACTIVE CURSES TUI & LIVE ANSWER SUBMISSION

Objective #1 (TCP 445 Flag): An nmap reason scan (`sudo nmap -p 445 --reason metasploit-2`) reveals response state `syn-ack`. Students who prefer a graphical dashboard can launch `cyberium tui` for a full-screen Tokyo-Night interface. Below illustrates the linear answering and instant toast feedback loop:

STEP 1: kali@box: ~/void/metasploit-2 (Curses TUI Live Lab Submission: syn-ack) TUI: Tokyo Night Palette

Apps Places Wed Oct 14 4:35 PM 80%

ssh

```
C Y B E  1/24 Solved (4%) | ★ EXPERT • 154 pts ◆ 83 coins a

[✓] #0  1. Download and impor…      ◆ QUESTION #1  |  ACTIVE
[⧖] #1  2. Scan the vulnerabl…
[ ] #2  3. Scan the vulnerabl…      2. Scan the vulnerable
[ ] #3  4. Check the Nmap hel…      Linux machine for the
[ ] #4  5. Scan the vulnerabl…      TCP port 445 using the
[ ] #5  6. Check the Nmap hel…      flag '--reason' and
[ ] #6  7. Scan your network …      find the displayed
[ ] #7  8. Run 'sudo nmap 8.8…      reason output.
[ ] #8  9. Run the scan 'sudo…
[ ] #9  10. Find the number o…
[ ] #10 11. Run the scan 'sud…
[ ] #11 12. Use the 'netmask …
[ ] #12 13. Run the command '…
[ ] #13 14. Find the Nmap fla…
[ ] #14 15. Run the scan 'sud…
[ ] #15 16. Scan the vulnerab…
[ ] #16 17. Run 'sudo nmap 8.…
[ ] #17 18. Scan the vulnerab…
[ ] #18 19. Find the Nmap fla…
[ ] #19 20. Find the Nmap fla…
[ ] #20 21. Find the Nmap fla…
[ ] #21 22. Run Netdiscover (…
[ ] #22 23. Download the Nmap…
[ ] #23 24. Analyze the conve…

❯ Submit Answer (LAB): syn-ack
```

```
13:34 ~/void/metasploit-2  ◆ metasploit-2
kali ❯ target
◆ Target Status
  ├─ Workspace : /home/kali/void/metasploit-2
  ├─ Target    : metasploit-2
  ├─ Hostname  : metasploit-2
  ├─ IP        : 172.30.30.124
  └─ Scope     : Single target

13:34 ~/void/metasploit-2  ◆ metasploit-2
kali ❯ sudo nmap -p 445 --reason metasploit-2
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-14 13:34 +00
00
Nmap scan report for metasploit-2 (172.30.30.124)
Host is up, received arp-response (0.00033s latency).

PORT    STATE SERVICE      REASON
445/tcp open  microsoft-ds syn-ack ttl 64
MAC Address: 52:54:00:2E:74:8D (QEMU virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.10 seconds

13:34 ~/void/metasploit-2  ◆ metasploit-2
kali ❯

13:34 ~/void/metasploit-2  ◆ metasploit-2
kali ❯
```

STEP 2: Interactive Curses TUI (Real-Time Toast Feedback & Checkmark Badges) [✓] +POINTS AWARDED

Apps Places Wed Oct 14 4:35 PM 80%

ssh

```
C Y B E R I U M | RTX2026 | ZX301 Penetration Testing › Scanning   2/24 Solved (8%) | ★ EXPERT • 154 pts ◆ 83 coins

[✓] #0  1. Download and import the vulnerable L…   ◆ QUESTION #2  |  ACTIVE OBJECTIVE [⧖]
[✓] #1  2. Scan the vulnerable Linux machine fo…
[⧖] #2  3. Scan the vulnerable Linux machine an…   3. Scan the vulnerable Linux machine and find the state of the TCP port
[ ] #3  4. Check the Nmap help menu and find th…   number 8782.
[ ] #4  5. Scan the vulnerable Linux machine an…
[ ] #5  6. Check the Nmap help menu and find th…
[ ] #6  7. Scan your network using the 'List Sc…
[ ] #7  8. Run 'sudo nmap 8.8.8.8 -sn', analyze…
[ ] #8  9. Run the scan 'sudo nmap 8.8.8.8 -p 5…
[ ] #9  10. Find the number of ports Nmap scans…
[ ] #10 11. Run the scan 'sudo nmap 192.168.1.1…
[ ] #11 12. Use the 'netmask -r <IP/Subnet>' co…
[ ] #12 13. Run the command 'netmask -r 192.168…
[ ] #13 14. Find the Nmap flag that disables DN…
[ ] #14 15. Run the scan 'sudo nmap 8.8.8.8 -PP…
[ ] #15 16. Scan the vulnerable Linux machine u…
[ ] #16 17. Run 'sudo nmap 8.8.8.8 --top-ports …
[ ] #17 18. Scan the vulnerable Linux machine u…
[ ] #18 19. Find the Nmap flag to save the outp…
[ ] #19 20. Find the Nmap flag to save the outp…
[ ] #20 21. Find the Nmap flag to save the outp…
[ ] #21 22. Run Netdiscover (netdiscover -r <yo…
[ ] #22 23. Download the Nmap XML file, convert…
[ ] #23 24. Analyze the converted HTML file and…

[✓] CORRECT! +Points awarded for #1.

[↑/↓] Nav [Enter] Answer [/] Search [n] Note [m] Modules [P] Tracks [x] Extra [t] Theme [b] Book [v] Video [p] Profile
```

FIGURE 6: Sequential Curses TUI execution: live in-terminal submission (syn-ack) followed by instant toast notification and unlocked badge.

📓 STUDENT SCRATCHPAD: IN-TERMINAL LAB MEMOS (cyberium note / notes)

Students frequently discover credentials, backdoors, hashes, and recon notes during exercises. Rather than juggling external text editors, students use `cyberium note "memo"` to link findings directly to the active question. `cyberium notes` prints a structured lab notebook.

kali@box: ~ (Student Field Notes: cyberium note & cyberium notes) PERSISTED: ~/.config/cyberium/notes.json

```
┌─ 14:22  ~
└─ kali ❱ cyberium note Loot: P:12345! U:Admin

[✓] Saved personal note for Question #83:
  Loot: P:12345! U:Admin


┌─ 14:22  ~
└─ kali ❱ cyberium notes

C Y B E R I U M [JOHN BRYCE] | ALL SAVED SCRATCHPAD NOTES (Post (Linux))
──────────────────────────────────────────────────────────────────────

#83
  Loot: P:12345! U:Admin

──────────────────────────────────────────────────────────────────────

Total notes: 1
```

kali@box: ~ (Student Telemetry: cyberium profile) ★ 8 GOLD TROPHIES

```
┌─ 14:16  ~
└─ kali ❱ cyberium profile

C Y B E R I U M [JOHN BRYCE] | OPERATOR DOSSIER (Student Profile)
──────────────────────────────────────────────────────────────────────

  Operator:  Denis Albin (s2)  |  Division: John Bryce (RTX2026)
  Rank:      ★ EXPERT  |  Status:  SPECTO ● LIVE
  Points:    154 pts  |  Coins:  83 coins  |  Certs: 7 verified

  Performance Benchmark (PBR):
    Offense: 97%  |  SOC Operations: 98%  |  Defense: 96%

  Trophies & Specializations:
    ★ 8 Gold Trophies  ★ 2 Silver  ★ 1 Bronze
──────────────────────────────────────────────────────────────────────


┌─ 14:16  ~
└─ kali ❱ _
```

● ACADEMIC PROPOSITION FOR JOHN BRYCE INSTRUCTORS & LECTURERS

1. Maximizes Student Focus
Keeps students immersed in their Kali terminal, eliminating browser tab-switching and workflow interruptions during hands-on exercises.

2. Authentic Industry Habits
Trains students to think and work like professional penetration testers, using CLI flags, automated scripting, and pipes.

3. Lightweight Deployment
Runs effortlessly across standard student environments: VMware Workstation, VirtualBox, WSL2, or remote jump boxes.

PROJECT AUTHOR: DENIS ALBIN (STUDENT CODE: S2)
John Bryce Academy | RTX2026 Offensive Cyber Warfare Division | Verified & Operational

[✓] VERIFIED 34/34 TESTS STATUS: PRODUCTION READY